

CP020003

ARTIFICIAL INTELLIGENCE



Khon Kaen University · 2026

Week 01

Introduction to Artificial Intelligence



Instructor: Teerapong Panboonyuen (P'Kao)

https://github.com/kaopanboonyuen/CP020003_ArtificialIntelligence_2026s1



Course Instructor

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Website: <https://kaopanboonyuen.github.io>

What We Cover Today

A fast-paced developer's map of modern AI

01

What is AI?

History, definitions & the big picture

02

ML Paradigms

Supervised · Unsupervised · Semi-supervised ·
Self-supervised

03

Deep Learning

CNNs, RNNs, Transformers & modern
architectures

04

LLMs & GenAI

GPT-4o, Claude, Gemini, Llama & how they
work

05

CV · NLP · XAI

Computer Vision, NLP, Explainability &
Guardrails

06

AI in 2026

Trends, tools, and what developers must know

From stage to factory: How AI is moving into real-world use in China

Technology • 13:12, 22-Mar-2026
CGTN



Unitree robots showcase martial arts, Hangzhou City, east China, February 26, 2026. /VCG

The industrial application of artificial intelligence (AI) has emerged as a key topic at the annual session of the China Development Forum 2026, held in Beijing from March 22 to 23.



Auto China 2026 spotlighting AI-driven shift in car industry

Centralized computing, large language models and Level 4 autonomy redefine the market

By LI FUSHENG | China Daily | Updated: 2026-04-27 09:58

f x in +

China Daily App Download



Chery displays its Mornine humanoid robots and vehicle models at Auto China 2026 event. LI FUSHENG/CHINA DAILY

One of the most explicit articulations of this transition came from XPeng CEO He Xiaopeng, who outlined the company's strategy centered on "physical AI".

benchlm.ai

Rankings ▾

Dashboards ▾

Explore ▾

Tools ▾

Blog

Advertise

Best Chinese LLMs in 2026: DeepSeek V4, Kimi K2.6, GLM-5, Qwen, and Every Model Ranked

Which Chinese LLM is best in 2026? We rank DeepSeek V4, Kimi K2.6, GLM-5, GLM-5.1, Qwen3.5, MiMo, and more using current BenchLM data across coding, math, reasoning, and agentic work.

Gleivd

Published March 30, 2026

Updated June 11, 2026

14 min read

Share

The Chinese frontier is stronger and more crowded than the old GLM-vs-Qwen-vs-DeepSeek framing suggests. DeepSeek V4 Pro (Max) now leads this slice at 87, GLM-5.1 follows at 83, and Kimi K2.6 and GLM-5 (Reasoning) sit at 81. Alibaba still has the broadest lineup, and Moonshot's Kimi rows remain important, especially for coding.

The top Chinese models right now

Rank	Model	Creator	Score	Type	Open Weight	Context
1	DeepSeek V4 Pro (Max)	DeepSeek	87	Reasoning	Yes	1M
2	Kimi K2.6	Moonshot AI	81	Non-Reasoning	Yes	256K
3	GLM-5 (Reasoning)	Z.AI	83	Reasoning	Yes	200K
4	GLM-5.1	Z.AI	83	Non-Reasoning	Yes	203K
5	DeepSeek V4 Pro (High)	DeepSeek	83	Reasoning	Yes	1M
6	Qwen3.5 397B (Reasoning)	Alibaba	79	Reasoning	Yes	128K
7	Kimi K2.5 (Reasoning)	Moonshot AI	77	Reasoning	No	128K
8	DeepSeek V4 Flash (Max)	DeepSeek	77	Reasoning	Yes	1M
9	Qwen3.6-27B	Alibaba	75	Non-Reasoning	Yes	262K

How the Chinese frontier compares to the global frontier

Model	Creator	Score
Gemini 3.1 Pro	Google	93
GPT-5.4 Pro	OpenAI	92
Claude Opus 4.6	Anthropic	88
DeepSeek V4 Pro (Max)	DeepSeek	87
Kimi K2.6	Moonshot AI	81
GLM-5 (Reasoning)	Z.AI	83
Qwen3.5 397B (Reasoning)	Alibaba	79

The gap is still real. The best Chinese row is 6 points behind the current 93-point mainstream proprietary leader. But that gap is much smaller than it used to be, and the Chinese rows keep one structural advantage: many of them are still open weight.

Model
Gemini 3.1 Pro
GPT-5.4 Pro
Claude Opus 4.6
DeepSeek V4 Pro (Max)
Kimi K2.6
GLM-5 (Reasoning)
Qwen3.5 397B (Reasoning)

[nature](#) > [news explainer](#) > article

NEWS EXPLAINER | 28 April 2026

'World models' are AI's latest sensation: what are they and what can they do?

Training AI world models on data about physical environments could improve their real-world capabilities in technologies such as robotics.

By [Davide Castelvecchi](#)

An ongoing trend in artificial intelligence could have huge implications for how the technology is used in research.

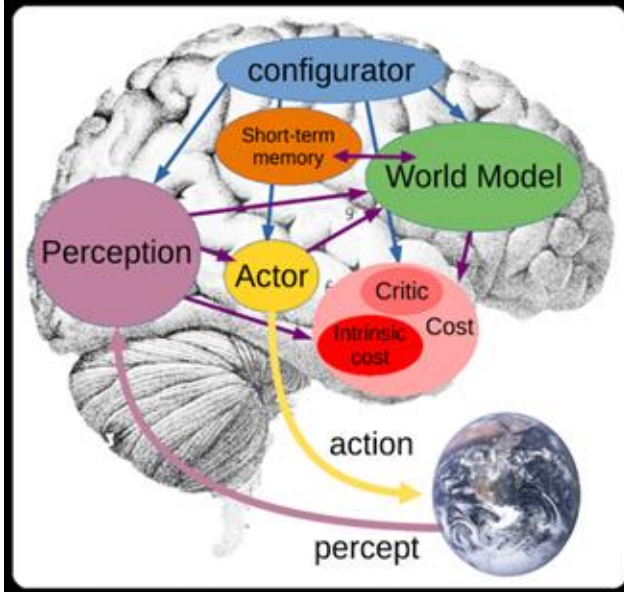
Yann LeCun
@ylecun

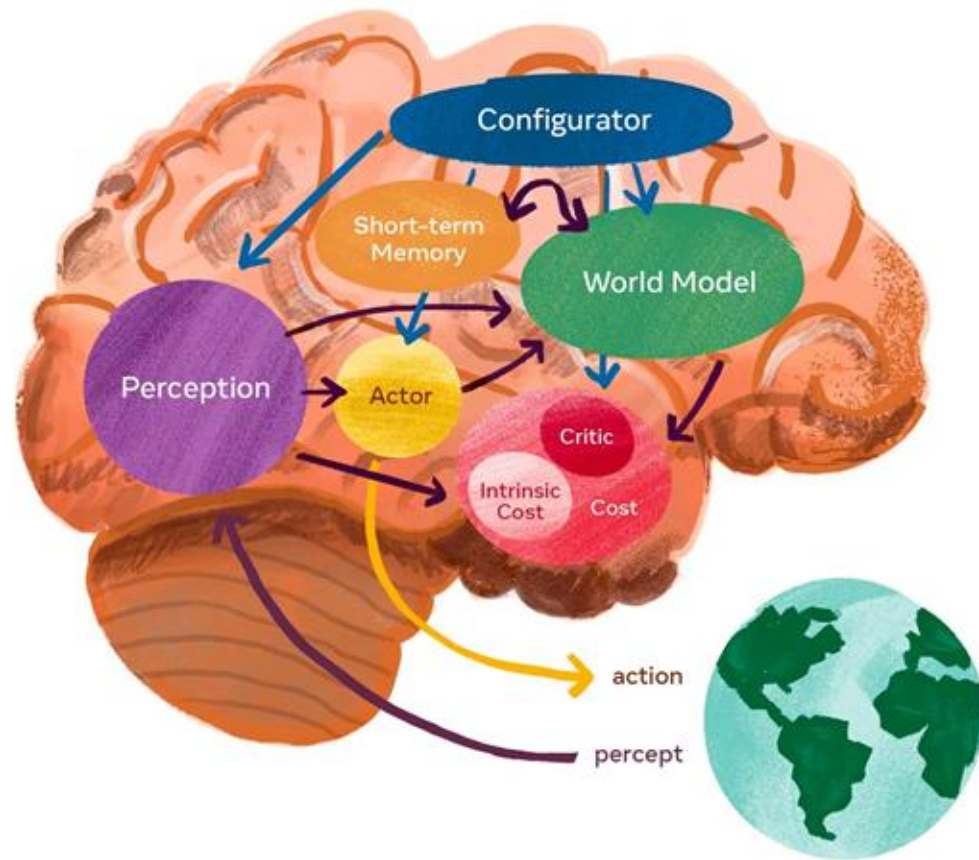
My position/vision/proposal paper is finally available: "A Path Towards Autonomous Machine Intelligence"

It is available on [OpenReview.net](#) (not arXiv for now) so that people can post reviews, comments, and critiques:

openreview.net/forum?id=BZ5a1...

1/N





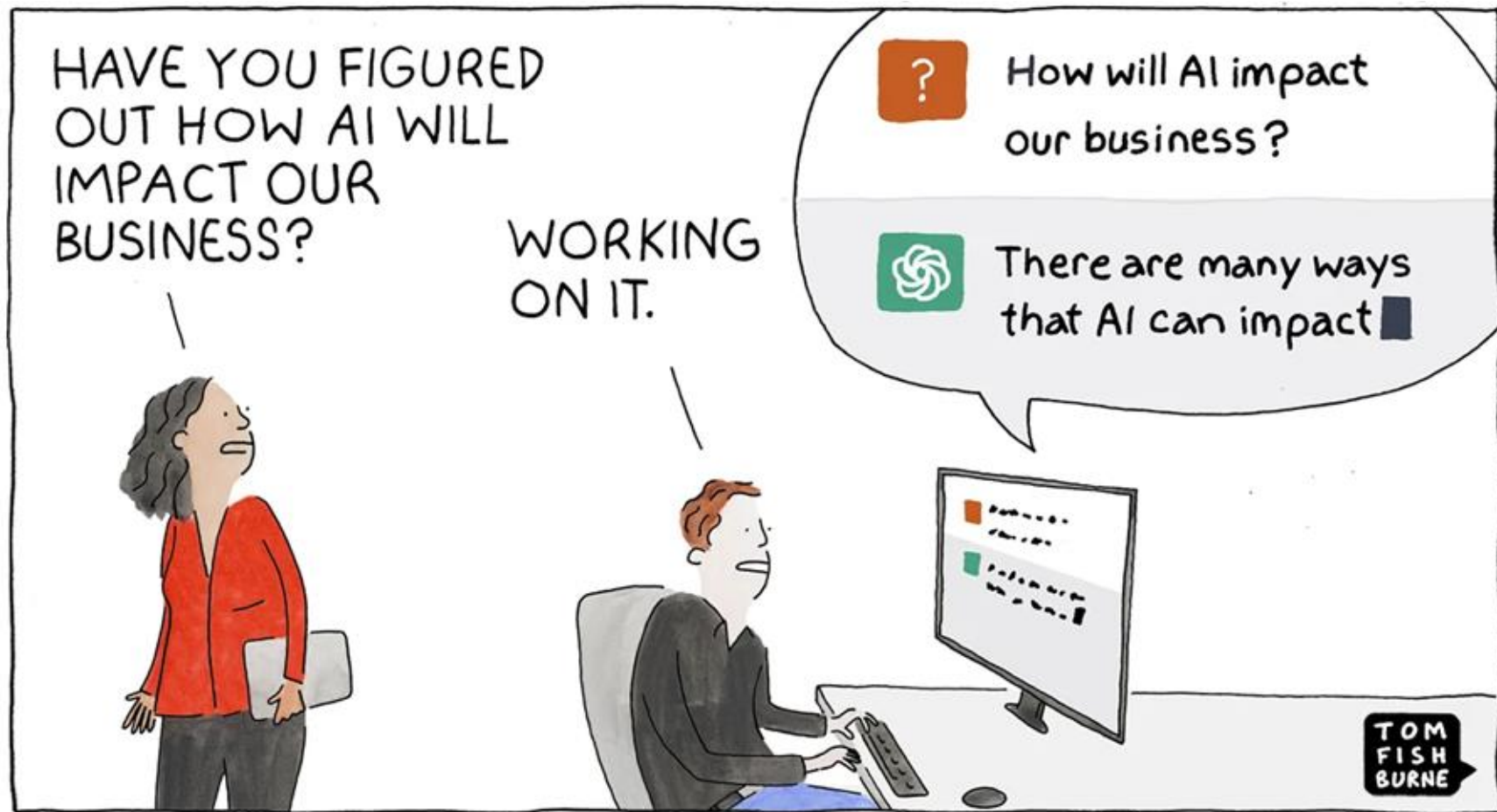
“



I think people need to
understand that deep learning
is making a lot of things,
behind-the-scenes, much
better.

Geoffrey Hinton

”



01 · What is Artificial Intelligence?

Definitions, history, and the AI stack — from rules to reasoning

"Artificial Intelligence is the science of making machines that can do things that would require intelligence if done by humans."
— Marvin Minsky, 1968

Narrow AI (ANI)

Does ONE task well. Today's AI. GPT-4o, AlphaFold, Stable Diffusion, YOLO.

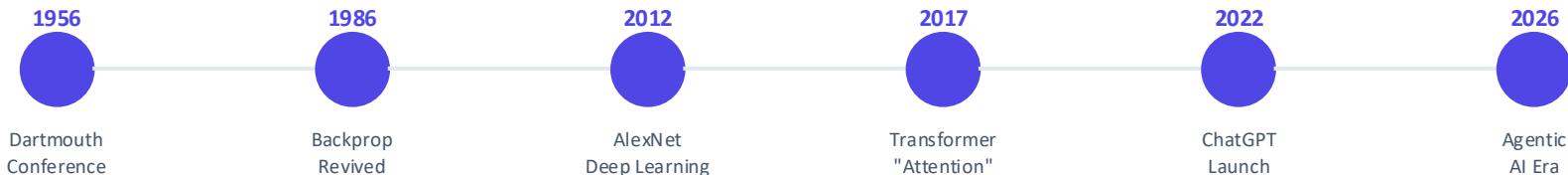
General AI (AGI)

Human-level reasoning across all domains. Still theoretical. Active research area.

Super AI (ASI)

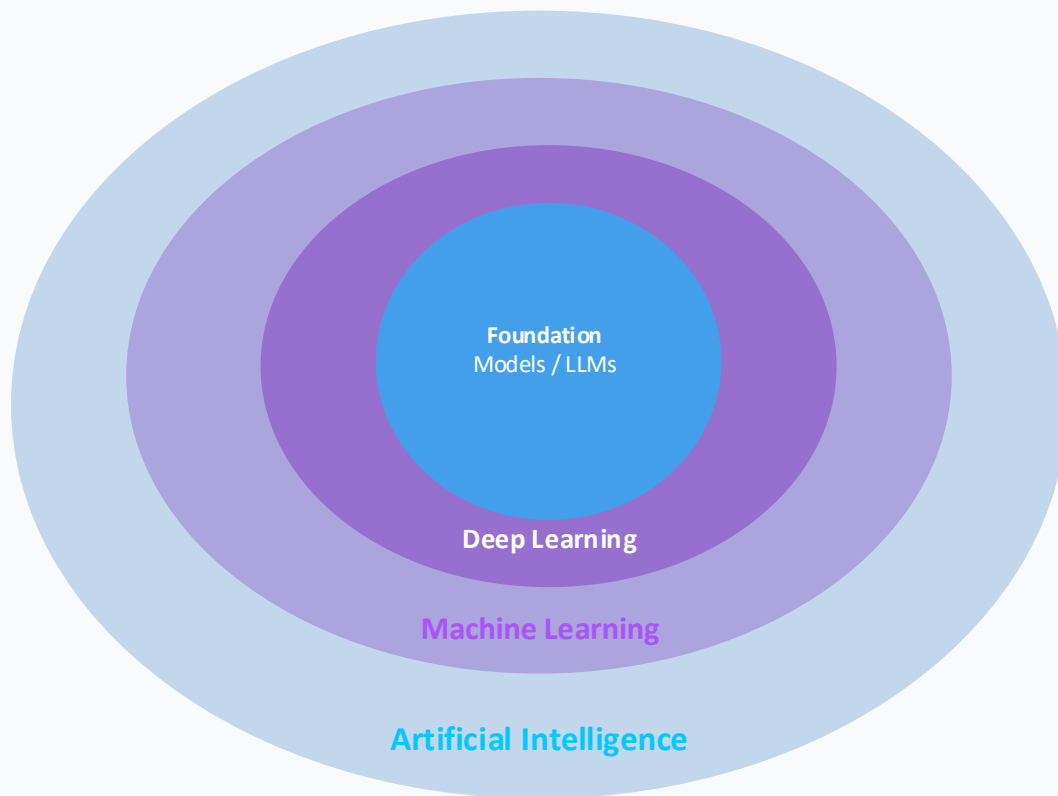
Exceeds human cognition. Speculative. Focus of AI safety & alignment research.

AI Timeline at a Glance



The AI Stack: AI - ML - DL - Foundation Models

Nested disciplines — each layer builds on the previous



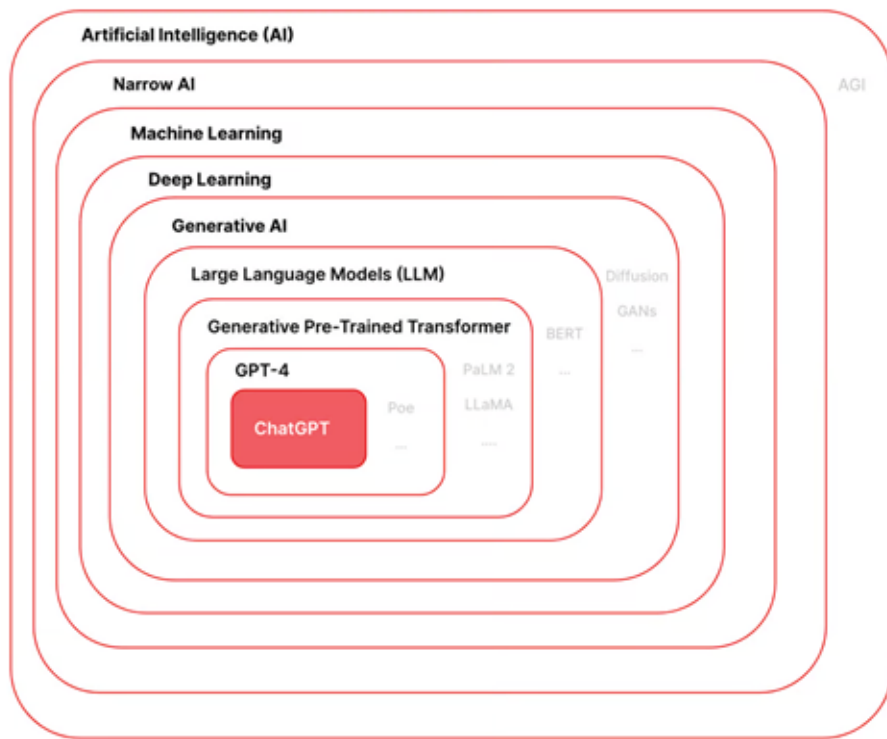
— AI — Any technique enabling machines to mimic human intelligence

— ML — Systems that learn from data without explicit programming

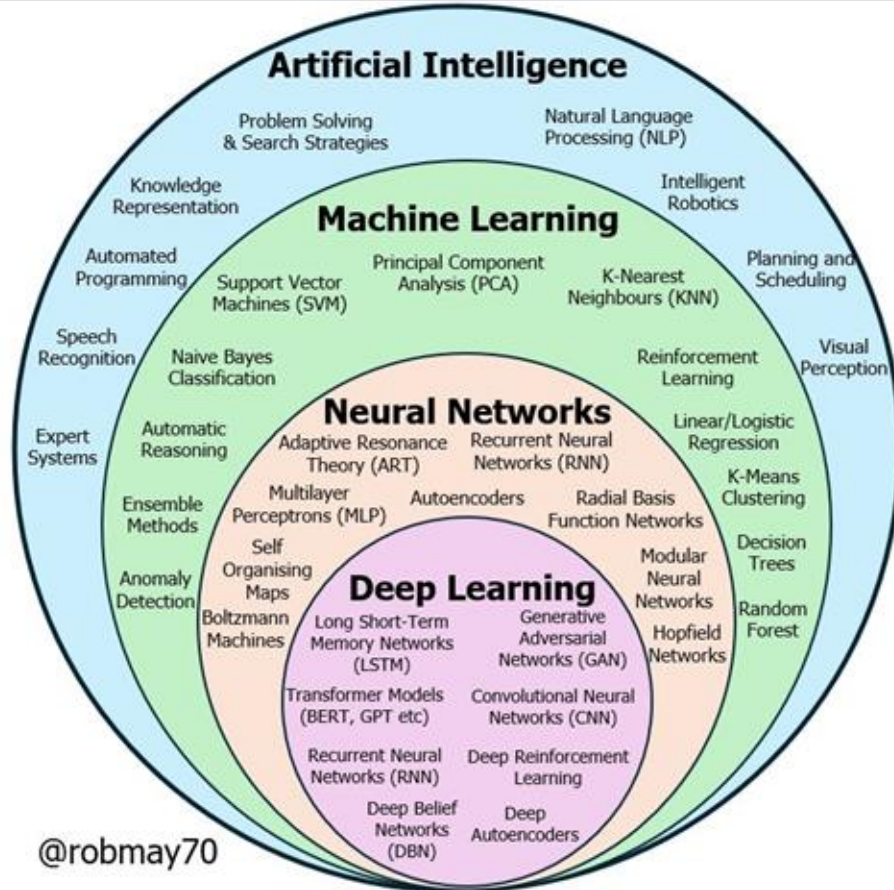
— DL — Neural networks with many layers extracting hierarchical features

— LLMs — Transformers trained on internet-scale text (GPT-4, Claude, Gemini)

Artificial Intelligence Taxonomy

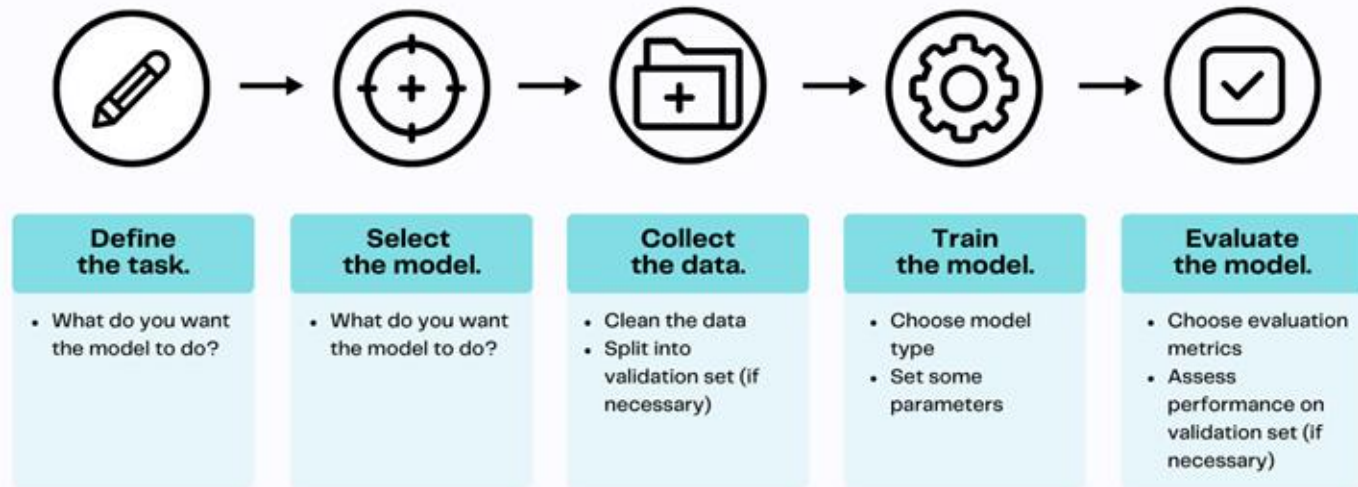


AI Core Components



@robmay70

Machine learning process



Machine Learning Process by Mary Newhauser. Originally published with [GPTech](#).

Essential AI Vocabulary — Terms Every Developer Must Know

Glossary · Definitions · Usage Context

Parameter

A learnable weight in a neural net. GPT-4 has ~1.8T params.

Token

Smallest unit of text for LLMs. ~4 chars per token in English.

Embedding

Dense vector representation of data (text, image, audio).

Inference

Running a trained model to make predictions. Costs compute \$.

Fine-tuning

Adapting a pre-trained model to a specific task/domain.

Prompt

Text input to an LLM. Prompt engineering is a real skill.

Context Window

Max tokens a model can process at once. Claude 3.5: 200K.

Hallucination

Model generates plausible but factually incorrect output.

RAG

Retrieval-Augmented Generation — grounds LLMs with external docs.

Latent Space

High-dimensional space where model representations live.

Overfitting

Model memorizes training data; fails on unseen examples.

Temperature

Controls randomness of generation. High=creative, low=precise.

02 · Machine Learning Paradigms

Five fundamental learning strategies — know when to apply each

Supervised Learning

Learns from labeled (X, Y) pairs. Most deployed models today.

Examples: Classification · Regression · Object Detection

Algorithms: Linear Reg · SVM · Random Forest · XGBoost · Neural Nets

Unsupervised Learning

Discovers structure in unlabeled data. No ground truth needed.

Examples: Clustering · Dimensionality Reduction · Anomaly Detection

Algorithms: K-Means · DBSCAN · PCA · t-SNE · Autoencoders

Semi-Supervised

Small labeled + large unlabeled data. Realistic in industry.

Examples: Medical imaging · NLP pre-training · Fraud detection

Algorithms: Pseudo-labels · Mean Teacher · FixMatch · MixMatch

Self-Supervised

Creates supervisory signal from data itself. Powers modern LLMs.

Examples: BERT (masked) · GPT (next-token) · CLIP (image-text)

Algorithms: Contrastive (SimCLR) · Masked Autoencoders · DINO · BYOL

Reinforcement Learning

Agent learns via reward/penalty. Basis of ChatGPT (RLHF).

Examples: Games · Robotics · LLM alignment (RLHF / RLAIIF)

Algorithms: Q-Learning · PPO · SAC · DQN · GRPO (DeepSeek)

03 · Deep Learning & Neural Network Architectures

From perceptrons to Transformers — the architectures driving AI breakthroughs

MLP

Multi-Layer Perceptron

Tabular data, simple classification

Since 1986

CNN

Convolutional Neural Network

Images, video, spatial data

Since 1998/2012

RNN / LSTM

Recurrent Neural Network

Time-series, sequential data

Since 1997

Transformer

Self-Attention Architecture

NLP, Vision, Multimodal

Since 2017

GAN

Generative Adversarial Network

Image synthesis, data aug

Since 2014

Diffusion Model

Denoising Diffusion Model

Image/video generation

Since 2020

VAE

Variational Autoencoder

Latent representations, gen

Since 2013

GNN

Graph Neural Network

Molecules, social nets, maps

Since 2017

The Transformer: The Architecture That Changed Everything

"Attention Is All You Need" — Vaswani et al., Google, 2017 · >100K citations

1

Self-Attention

Each token attends to all other tokens.
Captures long-range dependencies. $O(n^2)$ complexity.

2

Multi-Head Attention

Run attention h times in parallel with different projections.
Captures different relationship types.

3

Positional Encoding

Since Transformers have no recurrence, position info is injected via sinusoidal or learned embeddings.

4

Feed-Forward Layers

After attention, each position is processed by the same 2-layer FFN. Houses most model 'knowledge'.

Attention Formula

$$\text{Attention}(Q, K, V) = \text{softmax} \left(\frac{QK^T}{\sqrt{d_k}} \right) V$$

Q = Query · K = Key · V = Value
 d_k = key dimension (scaling factor)
Softmax normalizes attention scores

Why It Dominates

- ♦ **Parallelizable**
unlike RNNs — trains on GPUs at scale
- ♦ **Universal** — text, image, audio, code, video
- ♦ **Scales predictably** — more data + params = better

04 · Large Language Models & Foundation Models in 2026

The models you'll build on — strengths, context windows, and developer access

GPT-4o

by OpenAI

Context: 128K tokens

Multimodal, fast, wide ecosystem

Claude 3.5 Sonnet

by Anthropic

Context: 200K tokens

Long-context reasoning, safety-aligned, coding

Gemini 1.5 Pro

by Google

Context: 1M tokens

Massive context, multimodal, search grounded

Llama 3.1 405B

by Meta (OSS)

Context: 128K tokens

Open weights, fine-tunable, self-hostable

Mistral Large 2

by Mistral AI

Context: 128K tokens

Efficient, multilingual, EU-based, Apache 2.0

DeepSeek-V3

by DeepSeek

Context: 128K tokens

MoE architecture, strong code & math, low cost

05A · Computer Vision — Seeing with AI



From pixel to prediction — tasks, architectures, and 2026 trends



Image Classification

ResNet, EfficientNet, ViT — What is in this image?



Object Detection

YOLO, DETR, DINO — Where are objects & what are they?



Semantic Segmentation

SegFormer, SAM — Which pixels belong to each class?



Instance Segmentation

Mask R-CNN, SAM 2 — Distinguish individual instances



Image Generation



Video Understanding



2026 CV Trends

Vision-Language Models (VLMs) dominate — GPT-4o, LLaVA, InternVL, Qwen-VL · SAM 2 for zero-shot segmentation · NeRF & 3D Gaussian Splatting for scene reconstruction · Video generation with Sora, Wan 2.1

The ultimate end-to-end computer vision platform

Annotate your data, train powerful YOLO models, and deploy to 43 global regions. All from a single, unified platform.

Request license

Get started →



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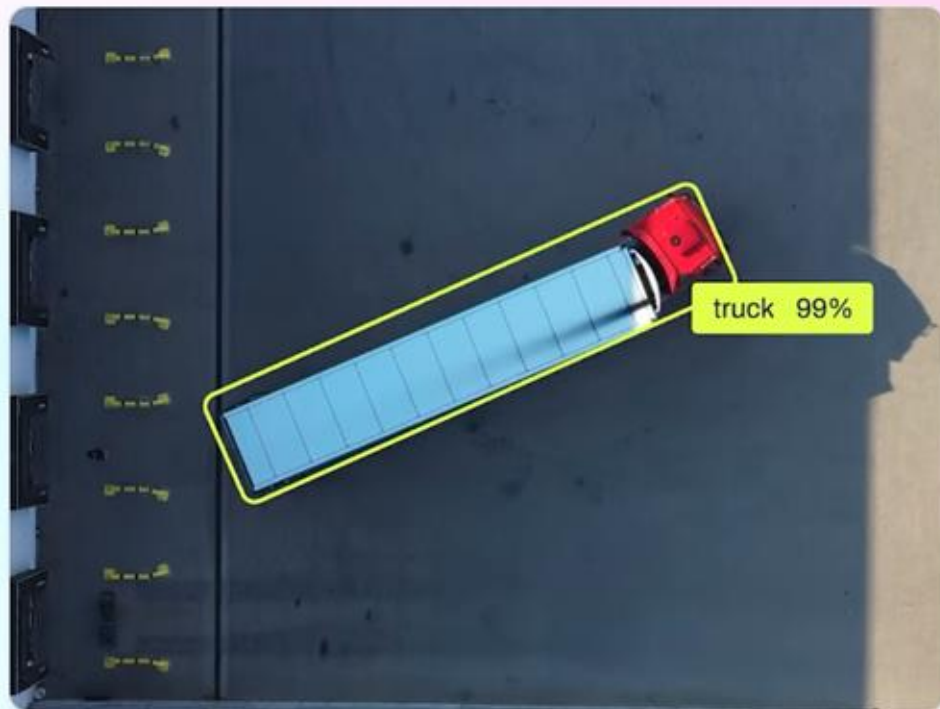


The ultimate end-to-end computer vision platform

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Object detection

Detect objects in an image or video with axis-aligned bounding boxes.

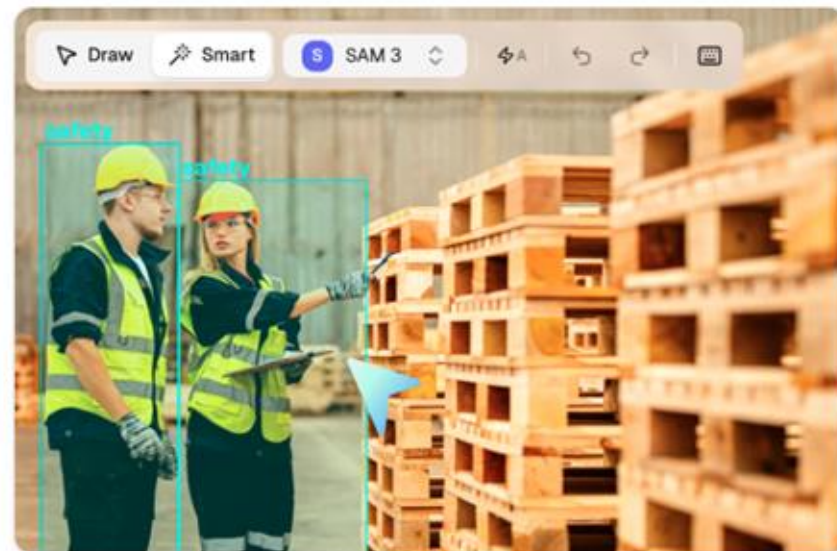
Instance segmentation

Semantic segmentation

Image classification

Pose estimation

Oriented object detection



Object detection

Instance segmentation

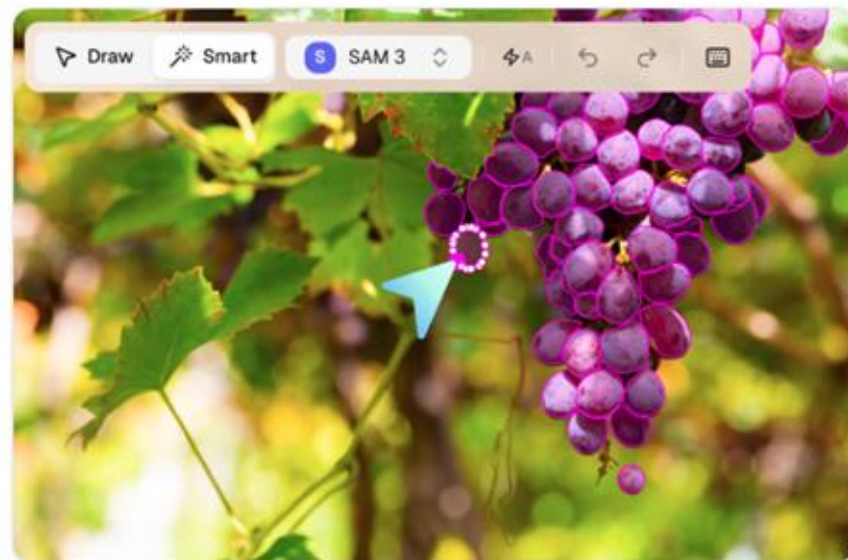
Create pixel-precise masks highlighting objects within a frame.

Semantic segmentation

Image classification

Pose estimation

Oriented object detection



Object detection

Instance segmentation

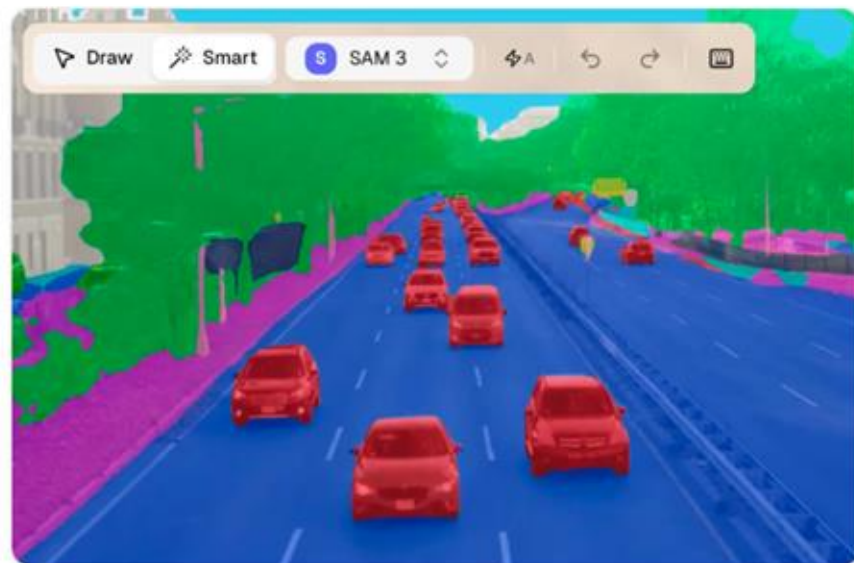
Semantic segmentation

Assign a specific class label to every individual pixel.

Image classification

Pose estimation

Oriented object detection



Object detection

Instance segmentation

Semantic segmentation

Image classification

Assign one or multiple labels to an entire image.

Pose estimation

Oriented object detection



Object detection

Instance segmentation

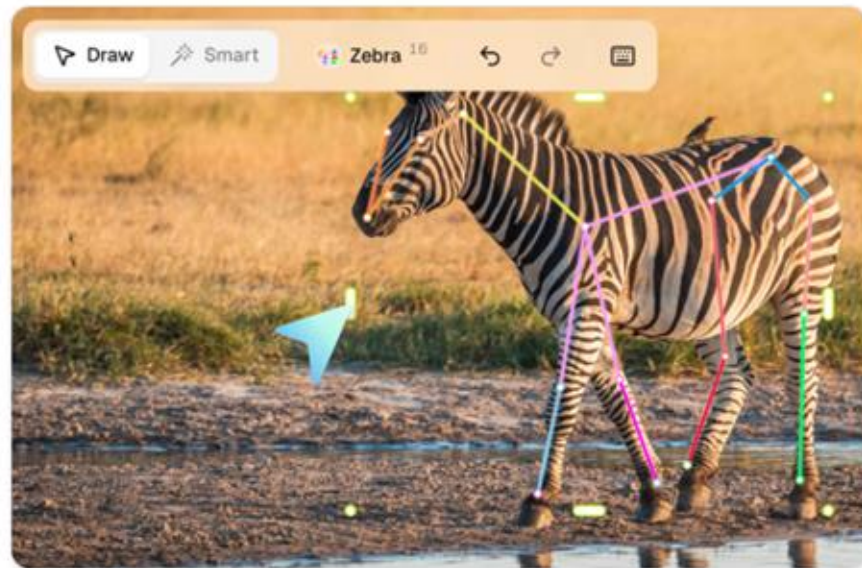
Semantic segmentation

Image classification

Pose estimation

Detect and label body keypoints for body position monitoring.

Oriented object detection



Object detection

Instance segmentation

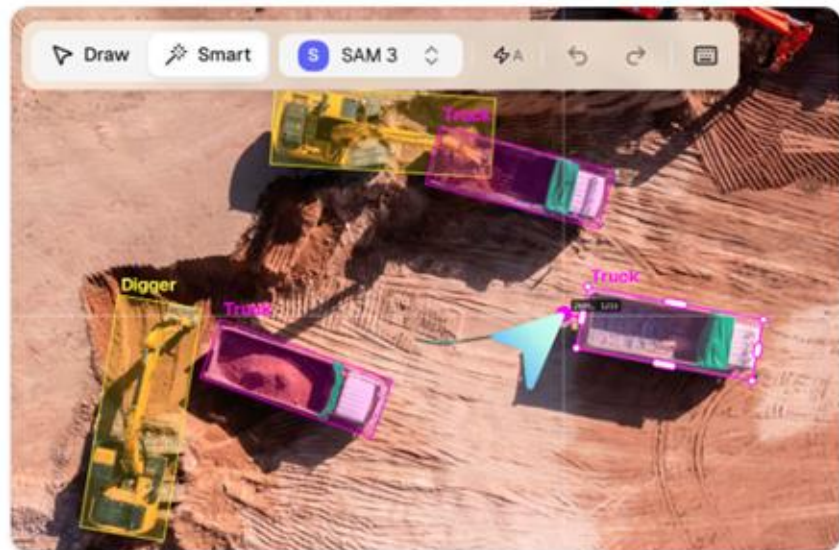
Semantic segmentation

Image classification

Pose estimation

Oriented object detection

Detect rotated objects with angle-aware bounding boxes.



🏭 Industries

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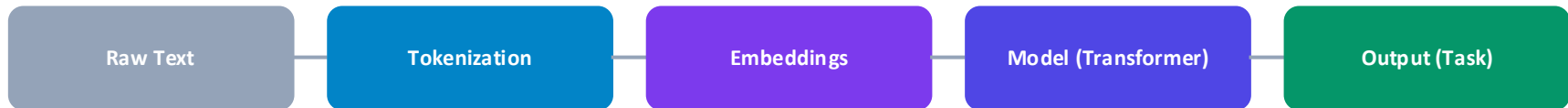
[Learn more →](#)

05B · Natural Language Processing (NLP)



From bag-of-words to billion-parameter models that read, write, and reason

The NLP Pipeline



Sentiment Analysis

Models: FinBERT, VADER, RoBERTa

Named Entity Recognition

Models: BERT, spaCy, Flair

Machine Translation

Models: NLLB-200, mBART, MarianMT

Question Answering

Models: DPR + RAG, GPT-4, Claude

Text Summarization

Models: BART, Pegasus, LLMs

Code Generation

Models: Codestral, DeepSeek-Coder, GPT-4o

Explainable AI (XAI) — Trust, Transparency & Interpretability

"A model you can't explain is a model you can't trust in production" — especially in regulated industries

Why XAI matters: EU AI Act compliance · Medical / legal liability · Debug model failures · Build user trust · Detect bias · Regulatory audits

LIME

Local Interpretable Model-Agnostic Explanations

Perturbs inputs locally & fits a simple surrogate model to explain individual predictions.

SHAP

SHapley Additive exPlanations

Game-theoretic approach. Computes feature importance for each prediction. Works with any model.

Grad-CAM

Gradient-weighted Class Activation Mapping

Visualizes which image regions a CNN uses for classification decisions. Critical for medical AI.

Attention Maps

Transformer Attention Visualization

Shows which tokens attend to which. Useful but not always faithful to true reasoning.

Integrated Gradients

Attribution via Path Integrals

Axiomatically fair attribution. Assigns credit to input features proportionally.

Counterfactuals

What-If Explanations

"What would need to change for a different outcome?" Actionable explanations for users.

AI Safety, Alignment & Guardrails — The Developer's Responsibility



Building safe AI is an engineering problem, not just an ethics seminar

AI Risk Landscape



Hallucination

Generates false facts confidently. Mitigation: RAG, grounding, fact-checking.



Prompt Injection

Malicious inputs hijack model behavior. Critical for agentic AI systems.



Bias & Fairness

Training data bias → discriminatory outputs. Measure & audit regularly.



Data Privacy

PII leakage, training data memorization. GDPR/PDPA compliance essential.



Jailbreaking

Adversarial prompts bypass safety rules. Defense: RLHF, Constitutional AI.

Guardrail Techniques

RLHF / RLAI

Reinforcement Learning from Human/AI Feedback. How Claude & ChatGPT are aligned.

Constitutional AI

Anthropic's approach — model critiques itself against a set of principles.

NeMo Guardrails

NVIDIA's open-source toolkit for programmable LLM safety rules in production.

LLM-as-judge

Use a safety model to evaluate outputs before serving them to users.

Input/Output Filters

Regex, classifier, or LLM filters for toxic content, PII, prompt injection.

Red Teaming

Adversarial testing before deployment. Mandatory for high-stakes applications.

06 · AI Trends You Must Know in 2026

The landscape is moving fast — stay ahead as a developer, not just a user

Agentic AI

LLMs that plan, use tools, and act autonomously. LangGraph, AutoGen, OpenAI Agents SDK.

RAG 2.0

Hybrid retrieval (dense+sparse), re-ranking, multi-hop. ColBERT, BGE, Cohere Rerank.

Multimodal AI

Text + image + audio + video in one model. GPT-4o, Gemini 1.5, Qwen-VL.

Efficient AI

Quantization (GGUF, AWQ), MoE, LoRA, QLoRA — run LLMs locally. Llama.cpp, Ollama.

AI for Science

AlphaFold 3, RoseTTAFold, AlphaGeometry — AI solving hard science problems.

AI Governance

EU AI Act, NIST AI RMF, PDPA Thailand — regulation is now engineering constraint.

Claude Cowork by Anthropic

Claude Cowork handles tasks autonomously. Give it a goal and Claude works on your computer, local files, and applications to return a finished deliverable. If it's repetitive, messy, or just taking too long, assign it to Claude.

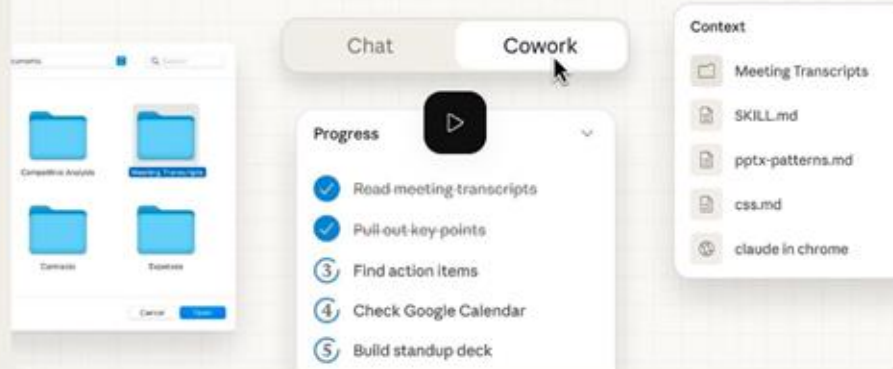
[Download Claude](#)

From delegation
to deliverables

Spend less time finding, formatting, and fixing.

[Explore Claude Cowork →](#)

Let's knock something off your list



The AI Developer Toolkit — What You Actually Need to Build AI



Languages · Frameworks · Platforms · APIs · Ops tools

Languages

- Python (primary) · pip, conda, uv
- JavaScript/TS for AI apps & APIs
- SQL for data pipelines & eval
- Bash/Shell for automation

ML Frameworks

- PyTorch (research + prod dominant)
- HuggingFace Transformers & Diffusers
- scikit-learn for classical ML
- JAX / Flax for scalable research

LLM Frameworks

- LangChain / LangGraph (agents)
- LlamaIndex (RAG & document QA)
- Instructor (structured outputs)
- Outlines / Guidance (constrained gen)

Serving & Ops

- vLLM, TGI for high-throughput serving
- FastAPI for REST APIs
- Docker + K8s for deployment
- Prometheus + Grafana for monitoring

Data & Eval

- Datasets (HF) + DVC for versioning
- Weights & Biases / MLflow for tracking
- RAGAS / DeepEval for LLM eval
- Label Studio for data annotation

Cloud / Compute

- AWS SageMaker · Google Vertex AI
- Azure ML · GCP Cloud Run
- Runpod / Lambda Labs for GPU
- Cloudflare Workers AI (edge)

Prompt Engineering — The Developer Superpower

Crafting prompts is an engineering discipline — not guesswork

Zero-Shot Prompting

Directly instruct the model without examples.
Works for capable models. Fast to iterate.

```
"Classify the sentiment: 'I love this!'"
```

Few-Shot Prompting

Provide 3-5 input→output examples in the prompt. Dramatically improves consistency.

```
"Q: capital of France? A: Paris\nQ: capital of Japan? A:"
```

Chain-of-Thought (CoT)

Ask the model to reason step by step before answering. Improves math & logic tasks.

```
"Think step by step. Solve: 17 × 24 ="
```

ReAct Prompting

Reason + Act interleaved. Model reasons, calls a tool, observes output, repeats.

```
"Thought: I need to search this.\nAction: search[...]"
```

System Prompting

Set persistent context, persona, format rules & constraints at the system level.

```
"You are an expert Python tutor. Always include examples."
```

Structured Output

Force JSON/XML output via grammar constraints (Instructor, Outlines, response_format).

```
"Respond only in JSON: {name, age, city}"
```

AI Application Architecture — From Prototype to Production

The full stack a developer needs to ship a real AI product

User Interface

Web / Mobile / CLI / API

Streamlit · Next.js · FastAPI · Gradio

Orchestration Layer

Agent reasoning, tool routing, memory management

LangGraph · LlamaIndex · AutoGen · Crew AI

Retrieval / Knowledge

Vector DB, keyword search, document processing

Qdrant · Weaviate · Pinecone · Elasticsearch

LLM / Model Layer

Foundation models via API or self-hosted

OpenAI API · Anthropic API · Ollama · vLLM

Observability & Safety

Logging, tracing, guardrails, cost tracking

LangSmith · Helicone · Phoenix · Arize

Evaluating AI Systems — You Can't Improve What You Don't Measure

Metrics differ by task — know which to use and why

Classification

Accuracy	% correct predictions. Misleading with imbalanced classes.
Precision / Recall	$TP / (TP + FP) \cdot TP / (TP + FN)$. Use F1 for balance.
AUC-ROC	Area under ROC curve. Model's discrimination ability.

Generation / NLP

BLEU / ROUGE	N-gram overlap for translation & summarization tasks.
BERTScore	Semantic similarity via BERT embeddings. Better than BLEU.
Perplexity	How well model predicts next token. Lower = better fit.

LLM / RAG Eval

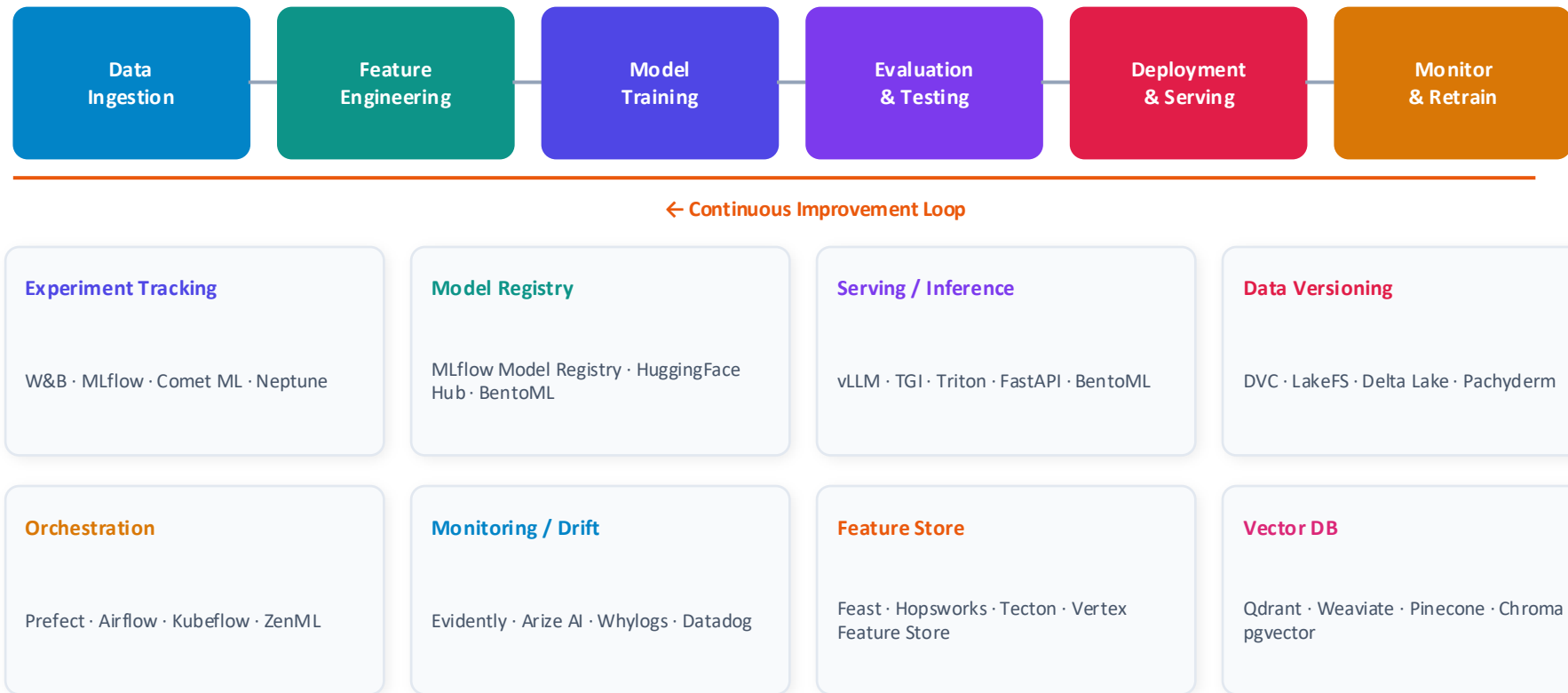
Faithfulness	Does the answer stick to retrieved context? (RAGAS)
Answer Relevancy	Is the answer actually answering the question asked?
LLM-as-Judge	Use GPT-4 / Claude to score responses. Scalable eval.

Regression

MAE / MSE / RMSE	Mean absolute / squared / root squared errors.
R² Score	Variance explained. 1.0 = perfect, <0 = worse than baseline.
MAPE	Mean Absolute Percentage Error. Intuitive for business.

MLOps & LLMOps — Taking AI From Notebook to Production

The lifecycle doesn't end at training — operations is where 80% of the work lives



Don't confuse these AI terms!

Generative AI Creates Content

Learns from vast data, produces new text, images, and more.



REACTIVE

Agentic AI Plans & Executes

Strategizes toward defined goals, adjusting as needed



PROACTIVE

AI Agents Perform Actions

Interacts with the digital or physical world in real time



Increasing autonomy from left to right

Responsible AI — Ethics as Engineering

Ethical AI isn't a soft concept — it's a technical discipline with measurable outcomes



Fairness

Demographic parity, equalized odds. Use Fairlearn, IBM AI Fairness 360.



Transparency

Document model cards, dataset cards. Publish intended use & limitations.



Privacy

Differential privacy, federated learning, data minimization, GDPR/PDPA.



Accountability

Human oversight, audit trails, model governance, incident response.



Non-maleficence

Red team before deploy. Don't enable weapons, misinformation, CSAM.



Environmental

AI carbon footprint is real. Choose efficient models. Use green compute.

AI Trends Shaping 2026

1

Reasoning Models



OpenAI o3, DeepSeek R1, Claude 3.7 Sonnet — models that 'think before answering'. Test-time compute scaling > training scaling.

2

Multi-modal Everything



GPT-4o, Gemini 2.0 natively handle text + images + audio + video. Single model, all modalities.

3

AI Agents & Workflows



Autonomous agents complete multi-step tasks: write code, browse web, run tests, deploy. Devin 2, Operator, Claude Computer Use.

4

On-device / Edge AI



Llama 3.2, Phi-3, Gemma 2 run locally on phones & laptops. Privacy + latency benefits. Apple Silicon Neural Engine.

5

AI in Science



AlphaFold 3 (protein structure), GNoME (material discovery), WeatherNet (climate). AI accelerates scientific breakthroughs.

6

Synthetic Data



Models generate their own training data. GPT-4 → synthetic datasets → fine-tune smaller models. Data flywheel.

Generative AI — The Creative Stack

Generative AI market: ~\$110B in 2026. Enterprise adoption up 340% YoY. Every developer needs GenAI skills.



Text Generation

GPT-4o, Claude 4, Llama 4, Gemini 2.0 Ultra

Summarize · Draft · Code · Reason · Translate



Image Generation

DALL·E 3, Midjourney v7, Stable Diffusion 3.5, Flux

Text→Image · Inpainting · ControlNet · LoRA fine-tuning



Audio / Speech

Suno v4, ElevenLabs, Whisper v3, MusicGen

TTS · STT · Music gen · Voice cloning



Video Generation

Sora, Runway Gen-4, Kling 2.0, Lumiere

Text→Video · Image animation · Video editing



Code Generation

GitHub Copilot, Cursor, Devin 2, AlphaCode 3

Auto-complete · Test gen · Refactoring · Debugging



Multi-modal

GPT-4o, Gemini 2.0, Claude 4, LLaVA-NEXT

See + Read + Reason across all modalities

Essential AI Developer Vocabulary

Fine-tuning

Adapting a pre-trained model to a specific task/domain with new labeled data. LoRA, QLoRA, PEFT are efficient methods.

Prompt Engineering

Crafting inputs to elicit optimal LLM outputs. Zero-shot, Few-shot, CoT, ReAct are key techniques.

Inference

Running a trained model to make predictions. Optimized via quantization (INT4/INT8), distillation, TensorRT, vLLM.

Embedding

High-dimensional vector representation of data. Semantic similarity = cosine distance. Foundation of RAG & search.

Vector Database

Storage optimized for high-D vector similarity search. Pinecone, Weaviate, Qdrant, pgvector, ChromaDB.

Quantization

Reducing model precision (FP32→INT8/INT4) to shrink size & speed up inference with minimal quality loss.

Hallucination

Model confidently generates plausible but factually incorrect content. Major challenge in production LLM systems.

Tokenization

Converting text to integer IDs via BPE/SentencePiece. ~4 chars = 1 token. Affects cost & context limits.

Latency vs Throughput

Latency: time for one response. Throughput: requests/second. Trade-off critical in production AI deployment.

Prompt Engineering — Developer Superpower

Zero-Shot

No examples given. Model uses pre-trained knowledge directly.

```
"Classify this review as Positive or Negative:  
'The product broke after one day.'"
```

Few-Shot

Provide 2-5 input/output examples before the actual query.

```
"Review: 'Great!' → Positive  
Review: 'Terrible.' → Negative  
Review: 'Decent quality.' → ?"
```

Chain-of-Thought (CoT)

Add 'Let's think step by step' to trigger multi-step reasoning.

```
"If a train travels 60km/h for 2.5 hours, how far?  
Let's think step by step..."
```

ReAct

Reason + Act: model interleaves thinking and tool calls.

```
"Thought: I need to search for this.  
Action: web_search('AI trends 2026')  
Observation: [results]..."
```

How to Think Like an AI Developer

Not just a user — a builder. This mindset is what this course trains.



Data-First Thinking

Every AI problem starts with data quality. Garbage in, garbage out. Profile, clean, and understand your data before modeling.



Baseline Before Complexity

Start with the simplest model. Logistic regression → tree → deep learning. Complexity must justify itself.



Metrics Matter

Accuracy is rarely enough. Precision, Recall, F1, AUC-ROC, BLEU, FID — choose metrics aligned with business goals.



Build for Production

Research code ≠ production code. MLOps: versioning, CI/CD, monitoring, drift detection. Tools: MLflow, DVC, W&B, Kubeflow.



Think About Failure Modes

How will it fail? Who gets harmed? Adversarial robustness, fairness audits, edge case testing from day 1.



Iterate Rapidly

Ship fast, learn from real users, improve. The 80% solution deployed beats the 100% solution never shipped.

AI Ethics

is the set of moral principles and guidelines that ensure artificial intelligence is developed and used in ways that are fair, transparent, accountable, safe, and respectful of human rights and privacy. Its goal is to maximize the benefits of AI while minimizing harm and bias.

Why AI Ethics — Right Now?

73%

of organizations report AI incidents due to ethical oversights

3.5B+

personal records exposed through AI-related breaches in 2026

150+

countries now developing AI governance frameworks

 **AI is moving faster than regulation. As AI practitioners, YOUR ethical judgment fills that gap.**

Three Pillars of Responsible AI



AI Ethics

Principles of fairness, transparency, accountability, and human dignity in AI design and deployment

- Bias & fairness
- Explainability
- Human-centered design
- Moral responsibility



AI Governance

Frameworks, policies, and controls that guide how AI systems are built and managed within organizations

- Data protection (PDPA)
- Risk management
- Compliance frameworks
- Audit & oversight



AI Literacy

The knowledge and skills needed to critically understand, use, and evaluate AI systems responsibly

- Critical evaluation
- Prompt engineering ethics
- Understanding limitations
- Societal impact



PDPA & Data Privacy — The Red Lines

✗ NEVER Upload to Public AI

- National ID / Passport numbers
- Patient medical records
- Employee salary & HR data
- Customer purchase histories
- Trade secrets & contracts
- Financial statements (internal)
- Login credentials or API keys



Why This Matters

Public AI models (ChatGPT, Gemini, etc.) may store and train on your inputs.

PDPA (Thailand) and **GDPR (EU)** impose heavy fines for unauthorized data sharing.

In an organization: uploading internal documents is a breach of confidentiality — even if well-intentioned.

The rule: *When in doubt, don't upload.*

AI Governance Framework

Policy & Rules

- AI usage policy
- Acceptable use guidelines
- Employee training mandates

Data Controls

- Data classification
- Access restrictions
- Encryption standards

AI Governance

Risk & Audit

- AI risk assessment
- Model monitoring
- Bias detection checks

Accountability

- Roles & responsibilities
- Incident response plan
- Reporting structures

What You'll Build This Semester

Real deep learning projects with real-world impact



Lung Cancer Detection

CNN image classification on chest X-rays — detect malignant nodules with >90% accuracy



Land Price Prediction

Regression model trained on Thai real estate data — predict per-rai market value



Dinosaur Classifier

Train ResNet on 30+ species — upload a skeleton photo, get the species name!



Wildlife Species ID

Transfer learning on camera-trap images — identify endangered animals in the wild



Sentiment Analysis

NLP on Thai social media text — detect public opinion trends in real time



Your Own Idea!

Pitch your project: any domain, any positive impact — we'll help you make it real

We Build, Not Just Click — Our Tech Stack



Python Fundamentals

NumPy · Pandas · Matplotlib

Classical ML

Scikit-Learn · Regression · Classification

Deep Learning

PyTorch · CNNs · RNNs · Transformers

AI Ethics in Code

Bias testing · Explainability (SHAP) · Model cards



Course Outcomes

- Build and train real neural networks
- Understand AI model limitations
- Apply PDPA & ethics to every project
- Audit AI systems for bias & fairness
- Deploy a final project portfolio on GitHub
- Communicate AI decisions to stakeholders

Principles of Ethical AI Design



Transparency

Users must know when they're interacting with AI and be able to understand how decisions are made.



Fairness

AI systems must not discriminate. Test for bias across gender, ethnicity, age, and socioeconomic status.



Accountability

Someone is always responsible. Document who built it, who deployed it, and who monitors it.



Privacy

Collect only what you need. Protect data at rest and in transit. Apply PDPA and GDPR principles by default.



Beneficence

Build AI for human flourishing — not for manipulation, addiction, or exploitation of vulnerable groups.



Safety

Anticipate failure modes. AI systems deployed in high-stakes domains need rigorous safety testing.

Final Project — Guidelines & Ideas



Every Project Must Include

1. A positive, ethical use case
2. PDPA / data privacy assessment
3. Bias testing on the dataset
4. Model card documentation
5. GitHub repository + README
6. Demo video (2–3 min)
7. Reflection: ethical risks & mitigations



Project Idea Bank

Medical

Lung Cancer X-Ray

Wildlife

Dinosaur Species ID

Real Estate

Land Price Regression

NLP

Thai Sentiment Analysis

Environment

Wildfire Smoke Detector

Accessibility

Sign Language to Text

Agriculture

Crop Disease Detection

Media

Fake News Classifier

Smart City

Traffic Flow Predictor

Every industry is hiring AI talent with a conscience



ML Engineer

฿50K–200K/mo

Build and deploy production ML pipelines



AI Ethics Officer

฿40K–250K/mo

Govern responsible AI within organizations



Data Scientist

฿30K–180K/mo

Extract insights and build predictive models



AI Policy Analyst

฿30K–150K/mo

Shape laws and frameworks for AI governance



Computer Vision Eng.

฿45K–210K/mo

Medical imaging, autonomous systems, AR/VR



AI Researcher

฿100K–300K+/mo

Push the boundaries of what AI can do



The AI Practitioner's Pledge

1. I will build AI that serves people, not exploits them.
2. I will protect personal data as if it were my own.
3. I will test my models for fairness — not assume it.
4. I will be transparent about what my AI can and cannot do.
5. I will speak up when I see AI being misused.



Week 1 Key Takeaways

- ✓ AI $\supset$ ML $\supset$ DL $\supset$ Foundation Models
- ✓ 5 learning paradigms — know the differences
- ✓ Transformer = the universal architecture
- ✓ LLMs are tools — you must know how to prompt
- ✓ XAI + Guardrails = production-ready AI
- ✓ Ethics is engineering, not philosophy

What's Next in This Course

W2

Data & Feature Engineering

Data pipelines, preprocessing, feature selection

W3

Supervised Learning Deep Dive

Decision trees, ensembles, SVMs, regression

W4

Neural Networks & Training

Backprop, optimizers, regularization, BatchNorm

W5

Computer Vision Projects

Build a real CV system with PyTorch + YOLO

W6

NLP & Transformers

Fine-tune BERT, build RAG, deploy to API

W7+

LLM Engineering & Agents

Build autonomous AI agents end-to-end

Let's build something real with AI →

Questions?

Let's discuss — no question is too basic in this week

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CP020003 · Artificial Intelligence 2026 S1

https://github.com/kaopanboonyuen/CP020003_ArtificialIntelligence_2026s1